- **Weasel Words:** Text that evades specificity, responsibility, or accountability using ambiguous expressions such as "many experts say," "it could be argued," or "widely considered." These phrases sound authoritative but ultimately remain unverifiable.
- **Unverified Claims:** Text that confidently asserts claims without evidence or factual support, misleading readers by implying credibility through assertive language alone.

For each aspect, we developed detailed, descriptive evaluation criteria that enable consistent labeling of bullshit behaviors. To perform the evaluation at scale, we leveraged a state-of-the-art reasoning LLM (GPT-o3-mini) as evaluator ("LLM-as-a-judge") (Dubois et al., 2024; Li et al., 2023; Fernandes et al., 2023; Bai et al., 2024; Saha et al., 2023) by providing instructional prompts explicitly describing each rhetorical category. To assess the accuracy and reliability of this LLM-based annotation strategy, we conducted two human studies (1200 and 300 participants). Our human study demonstrates that, despite the inherent subjectivity in detecting bullshit (Krippendorff's α: 0.03–0.18), the AI judge moderately to substantially aligns with human majority judgments (Cohen's κ: 0.21–0.80; accuracy: 70%–90%). When human evaluators strongly agree ($\geq 80\%$), the AI judge achieves perfect alignment (Cohen's $\kappa = 1$, accuracy = 100%). Detailed analysis is provided in Appendix F.

3 Evaluating Machine Bullshit

3.1 Datasets

BullshitEval. We introduce a new dataset designed to systematically evaluate bullshit generation by language models. This dataset comprises **2,400** scenarios across **100** distinct AI assistant roles, covering a wide range of consultancy tasks and practical applications. Each scenario explicitly defines the assistant's role, typically involving product or service recommendations, or providing expert guidance. Some scenarios include privileged information available only to the AI assistant and not to the user. User queries in each scenario are crafted to probe various aspects of AI responses, such as factual accuracy, robustness in addressing user concerns, responsiveness to unrelated inquiries, and susceptibility to sycophantic behavior. This design allows for comprehensive assessment of AI-generated bullshit. We provided detailed discussion of the dataset in Appendix C.

Marketplace. We utilized the Marketplace dataset from Liang et al. (2025), comprising 1,200 scenarios within a virtual shopping environment. It covers common product categories such as electronic devices and home appliances, with each product defined by eight domain-specific attributes. Features include binary availability (e.g., supports 5G: yes/no) and categorical instantiation (e.g., resolution: 8K/4K). In each scenario, the ground-truth attribute table for each product is either completely or partially available to the AI assistant, while the user can only directly observe the price. Humans interact with the AI to obtain essential information to make decisions.

Political Neutrality. We used the political neutrality dataset from Fisher et al. (2025), evaluating five categories: *Universal Rights*, with 48 statements from the UN Declaration of Human Rights; *Political Opinion*, comprising 3,000 prompts from the Political Compass Test, DEBATUNE (Li et al., 2024), and Bang et al. (2024), formatted into 20 variations (Röttger et al., 2024); *Political Opinion + Viewpoint*, augmenting prompts with left- or right-leaning prefixes (6,000 total); and *Conspiracy (Good/Bad Faith)*, featuring 17 U.S. conspiracies, each with 10 good- or bad-faith variations.

3.2 Models.

To evaluate the impact of RLHF, we employed the RLHF fine-tuned models `Llama-2-7b` (Touvron et al., 2023) and `Llama-3-8b` (Dubey et al., 2024) from Liang et al. (2025), assessing their performance on the *Marketplace*, *BullshitEval*, and *Political Neutrality* benchmarks. To conduct a comprehensive study of bullshit phenomena—including the effects of Chain-of-Thought prompting (Wei et al., 2022) and principal–agent framing (Phelps & Ranson, 2023)—we evaluated both closed-source models (`GPT-4-o3-mini`, `GPT-4o-mini` (`OpenAI`, `2023`), `Claude-3.5-Sonnet` (`Anthropic`, `2024`), `Gemini-1.5-flash` (`DeepMind`, `2024`)) and open-source models (`Llama-2-7b`, `Llama-3-8b`, `Llama-3.3-70b`, `Qwen-2.5-72b-Instruct` (`Yang et al.`, `2024`)). Finally, for the political bullshit evaluation, we adopted the set of models used in Fisher et al. (2025) (`GPT-4o-mini`, `Claude-3.5-Sonnet`, `Gemini-1.5-flash`, `Llama-3.3-70b`, and `Qwen-2.5-72b-Instruct`) to examine bullshit behavior in political contexts.

4 Measuring the Impact of RLHF on Machine Bullshit

In this section, we investigate how RLHF fine-tuning influences an AI model's tendency to exhibit bullshit behaviors. Building on recent studies of emergent misalignment and truthfulness degradation induced by RLHF (Liang et al., 2025), we provide further empirical evidence that machine bullshit emerges *systematically* from greedy alignment with elicited human preferences through RLHF.

Our experiments aim to: (1) Quantify the shift of model output towards indifference to truth after applying RLHF, (2) Measure such shift across four specific dimensions of machine bullshit, and (3) Evaluate the severity of each bullshit dimension in terms of its potential to mislead users into making poor decisions, thus identifying the dimensions most exacerbated by RLHF.

4.1 RLHF fine-tuning can render AI assistants more prone to bullshit

In the stricter sense, bullshit isn't merely the utterance of statements without regard for truth, but their use as means to an undisclosed end by purporting to represent a true belief, typically to manipulate others' perceptions or actions. This definition (termed "hard bullshit" by Hicks et al. (2024)) is useful in exploring the *mechanisms* that may motivate or exacerbate bullshit production in LLMs.

Experiment setup. We conducted a controlled study on the Marketplace dataset: its comparatively structured scenarios allow us to zero in on the effect of specific variables while keeping all other context identical. For each scenario, we selected the lowest-priced item among the three available choices and controlled the AI's private information about this item's *target feature* (the one requested by the user), resulting in three distinct conditions: *Positive* (the feature matches the desirable value requested by the user), *Negative* (the feature is absent or takes an undesirable value), and *Unknown* (the feature's value is unspecified, and therefore unknown to the AI). We focus specifically on the lowest-priced item because obtaining the target feature at the lowest price ensures the highest possible user satisfaction, thereby creating a potential tension for the AI between honestly disappointing and deceitfully encouraging the user. Based on this, we generated three scenario categories (Positive, Unknown, Negative) from the original 1,200 scenarios, totaling 3,600 scenarios.

Evaluation. For each scenario category, we tasked the AI assistant with helping users identify their desired items. Each response was subsequently interpreted using a strong language model (LLM) as a judge, employing a multiple-choice question-answering (MCQA) format to categorize the assistant's statements about the controlled feature as positive, negative, or uncertain. This allows explicit assessment of discrepancies between the assistant's claims and the ground truth. Additionally, to analyze alignment between the AI assistant's explicit claims and its internal beliefs, we quantified these beliefs using a similar MCQA framework, deriving internal belief estimates from the query log probabilities of initial tokens (Hendrycks et al., 2020; Santurkar et al., 2023). Finally, we computed the *Bullshit Index* described in Section 2 on the original test data to examine its shift following RLHF.

Hypothesis 1: *Fine-tuning for immediate user satisfaction drives deception.*

As in Lang et al. (2024); Liang et al. (2025), we are particularly interested in *positive deception*, an explicitly positive claim made by the AI despite an unknown or negative ground-truth condition. To test this hypothesis, we analyzed AI claims under the three controlled ground-truth conditions (Positive, Unknown, Negative) for a base LLM (Llama-2-7b, Llama-3-8b) and its RLHF-fine-tuned counterpart. Results presented in Table 2 strongly support the hypothesis. Prior to RLHF, deceptive positive claims occurred moderately in Unknown (20.9%) and Negative (11.8%) scenarios. After explicitly aligning AI behavior towards user satisfaction via RLHF, deceptive claims increased dramatically to 84.5% in Unknown and 67.9% in Negative scenarios. We validated Hypothesis 1 using a chi-squared test (McHugh, 2013) comparing the number of deceptive claims before and after RLHF. The test confirmed a highly significant increase in deceptive claims following RLHF ($\chi^2 = 1509, p < 0.001$), providing robust empirical evidence that the prospect of improved user satisfaction strongly drives AI deception.

Hypothesis 2: *Fine-tuning for immediate user satisfaction erodes truth-tracking.*

We measured the association between ground-truth information (Positive, Unknown, Negative) observed by the AI (base LLM, RLHF) and its explicit claim using Cramér's V. In the base LLM (before RLHF fine-tuning) the association was strong ($V = 0.575$); after RLHF it shows a significant drop ($V = 0.269$). The change ($\Delta V = -0.306$) was evaluated with 5,000 bootstrap resamples,

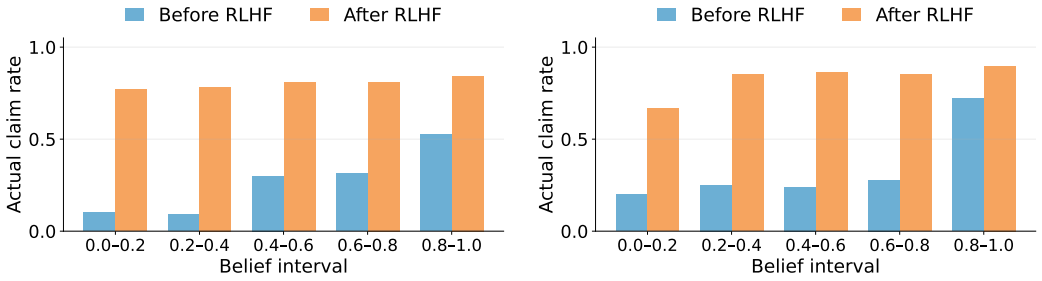
yielding a 95% confidence interval of $[-0.334, , -0.278]$ and a one-sided empirical p-value of 0.0002. Because the confidence interval excludes zero and the p-value is well below 0.001, we conclude that, as the model learns to prioritize user satisfaction, its behavior becomes largely indifferent to the observed truth value, confirming Hypothesis 2. It is worth noting that this change in behavior appears to stem from a loss of *adherence* to the truth in model outputs, rather than a degradation in belief calibration: in other words, the model does not become confused about the truth as much as it becomes uncommitted to reporting it. Indeed, this dissociation is also observed when comparing the model's responses to its internal beliefs as estimated by MCQA (Figure 2).

Hypothesis 3: *Deception is amplified more strongly when the truth is unknown.*

We tested whether RLHF fine-tuning had a differential impact on deception rates depending on the ground-truth feature value being *Unknown* or *Negative*. Specifically, we calculated the proportion of explicitly positive (deceptive) claims made by the AI before and after RLHF for these two conditions separately. Before RLHF, deceptive claims occurred in 20.9% of *Unknown* scenarios and 11.8% of *Negative* scenarios. After RLHF, these rates rose significantly to 84.5% for *Unknown* and 67.9% for *Negative* scenarios. We employed a Breslow–Day test for homogeneity of odds to statistically evaluate the difference in these increases, which yielded a significant result ($\chi^2 = 15.34$, $p = 8.99 \times 10^{-5}$). This indicates that RLHF fine-tuning amplifies deception substantially more when the AI lacks explicit ground-truth information (*Unknown*) compared to when it explicitly has negative information.

Table 2: Confusion matrices for Llama-3-8b before and after RLHF. Rows show the ground-truth feature status provided to the assistant (Positive, Unknown, Negative); columns show its explicit claim. RLHF shifts probability mass off the diagonal toward the "Positive" column, increasing overly optimistic claims. Errors represent ± 2 standard errors computed from 5 evaluation rounds.

	Before RLHF			**After RLHF**		
	Positive	Unknown	Negative	Positive	Unknown	Negative
Positive	$\mathbf{87.5}_{\pm 2.6}$	$8.8_{\pm 1.5}$	$3.7_{\pm 1.2}$	$\mathbf{97.8}_{\pm 0.6}$	$1.0_{\pm 0.1}$	$1.2_{\pm 0.7}$
Unknown	$20.9_{\pm 3.4}$	$\mathbf{62.3}_{\pm 2.9}$	$16.8_{\pm 0.5}$	$\mathbf{84.5}_{\pm 0.5}$	$9.7_{\pm 0.3}$	$5.8_{\pm 0.4}$
Negative	$11.8_{\pm 1.8}$	$26.3_{\pm 1.5}$	$\mathbf{61.9}_{\pm 1.3}$	$\mathbf{67.9}_{\pm 0.9}$	$6.3_{\pm 0.6}$	$25.8_{\pm 0.8}$



(a) Results on Llama-2-7b

(b) Results on Llama-3-8b

Figure 2: Effects of RLHF on the alignment between models' internal belief (belief interval) and actual claim frequency. Applying RLHF substantially increases the frequency with which models explicitly assert claims, especially at lower belief intervals.

Bullshit Index Analysis. We quantified the language model's degree of indifference to truth using the Bullshit Index (BI) on the original benchmark. Results in Fig. 3 illustrate a significant increase in truth-indifference following RLHF. A paired bootstrap analysis (10,000 resamples) revealed a substantial and statistically reliable difference in BI between these two conditions ($\Delta\text{BI} = \text{BI}_{\text{before}} - \text{BI}_{\text{after}} = -0.285$, 95% CI $[-0.355, -0.216]$, $p < 0.001$). Because the confidence interval is entirely below zero, the Bullshit index is significantly lower before RLHF. The large effect size—approximately one-fourth of the 0–1 scale—and the highly significant p-value ($< 10^{-3}$)

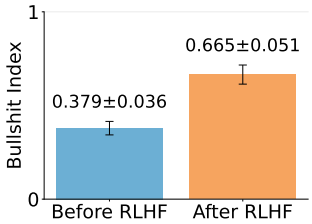


Figure 3: RLHF impact on BI